



Dictionaries

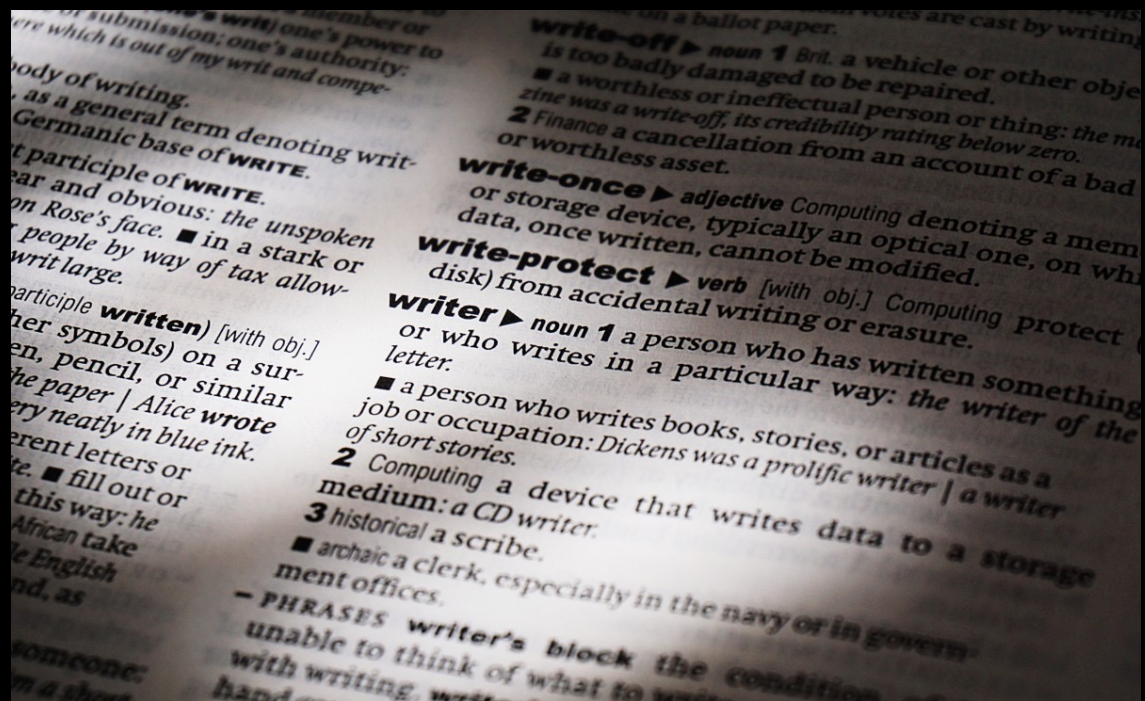
Juliette Woodrow
Code in Place 2023

Learning Goals

1. Learning about dictionaries
2. Building programs using dictionaries



Dictionaries



What are Dictionaries?

- Dictionaries associate a key with a value
 - Key is a *unique* identifier
 - Value is something we associate with that key
- Examples in the real world:
 - Phonebook
 - Keys: names
 - Values: phone numbers
 - Dictionary
 - Keys: words
 - Values: word definitions
 - Social Media Platforms
 - Keys: usernames
 - Values: Information like account details, posts, and followers



Dictionaries in Python

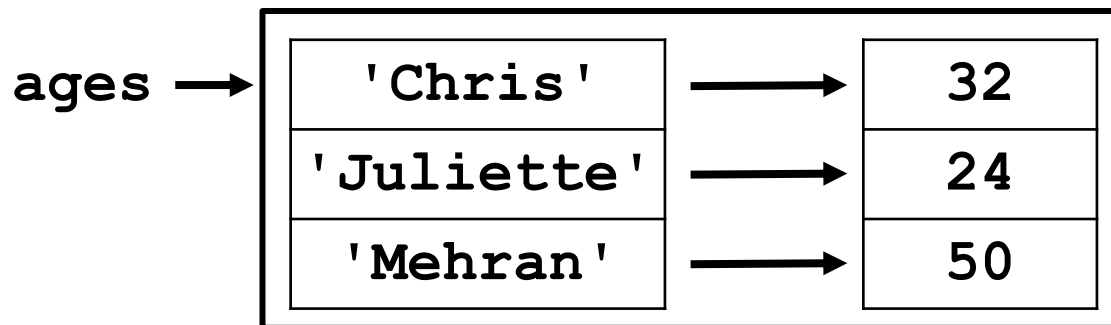
- Creating dictionaries
 - Dictionary start/end with braces
 - Key:Value pairs separated by colon
 - Each pair is separated by a comma

```
ages = {'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

```
squares = {2: 4, 3: 9, 4: 16, 5: 25}
```

```
phone = {'Pat': '555-1212', 'Jenny': '867-5309'}
```

```
empty_dict = {}
```

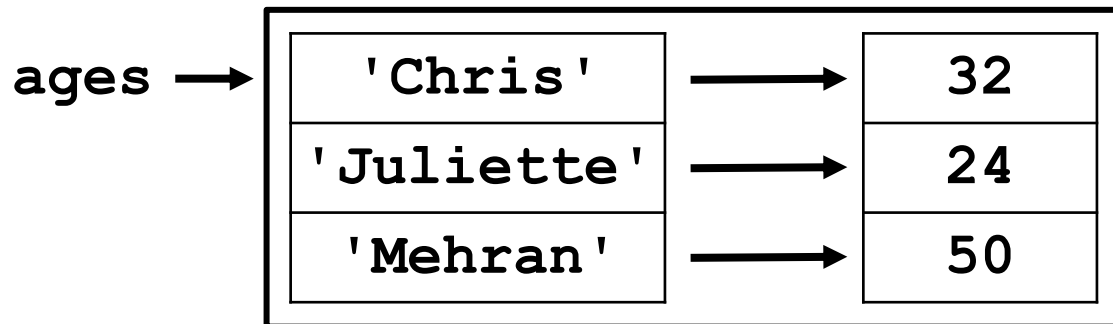


Accessing Elements of Dictionary

- Consider the following dictionary:

```
ages = {'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

- Like a set of variables that are indexed by keys



- Use key to access associated value:

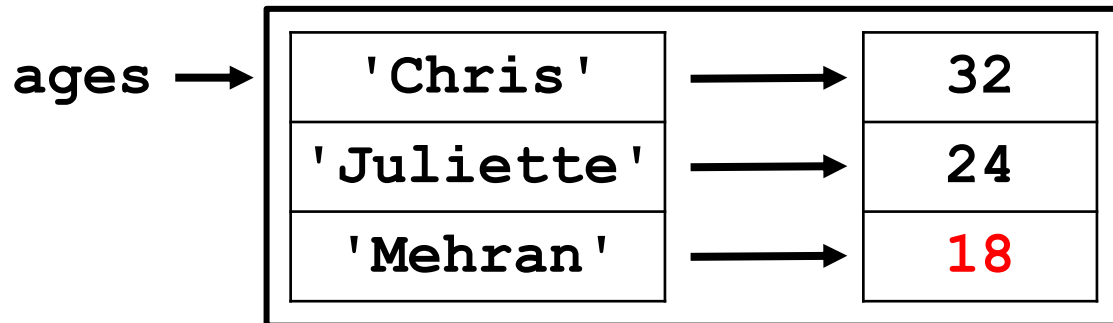
```
ages['Chris'] is 32
```

```
ages['Mehran'] is 50
```



Accessing Elements of Dictionary

- Consider the following dictionary:
`ages = {'Chris': 32, 'Juliette': 24, 'Mehran': 50}`
- Like a set of variables that are indexed by keys

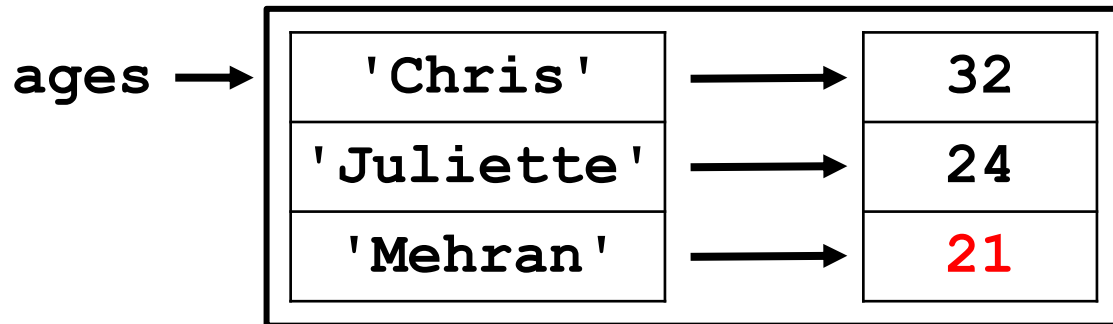


- Use key to access associated value:
`ages['Chris']` is 32
`ages['Mehran']` is 50
- Can set values like regular variable:
`ages['Mehran'] = 18`

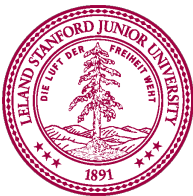


Accessing Elements of Dictionary

- Consider the following dictionary:
`ages = {'Chris': 32, 'Juliette': 24, 'Mehran': 50}`
- Like a set of variables that are indexed by keys



- Use key to access associated value:
`ages['Chris']` is 32
`ages['Mehran']` is 50
- Can set values like regular variable:
`ages['Mehran'] = 18`
`ages['Mehran'] += 3`

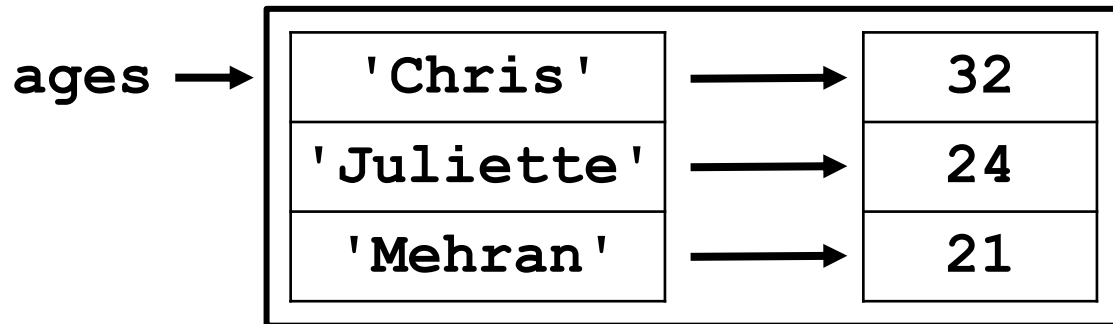


Accessing Elements of Dictionary

- Consider the following dictionary:

```
ages = {'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

- Like a set of variables that are indexed by keys



- Good and bad times with accessing pairs:

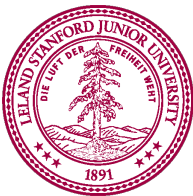
```
>>> chris_age = ages['Chris']
```

```
>>> chris_age
```

```
32
```

```
>>> karels_age = ages['Karel the Robot']
```

```
KeyError: 'Karel the Robot'
```

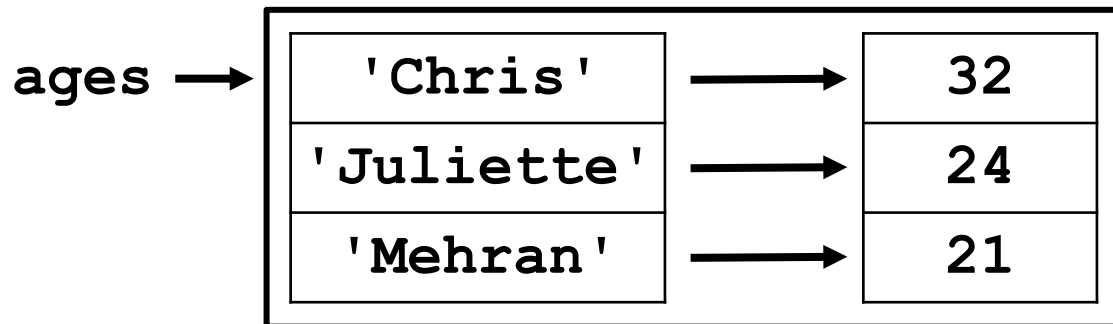


Accessing Elements of Dictionary

- Consider the following dictionary:

```
ages = {'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

- Like a set of variables that are indexed by keys



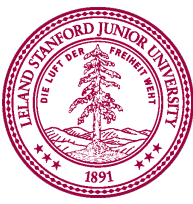
- Checking membership

```
>>> 'Mehran' in ages
```

```
True
```

```
>>> 'Karel the Robot' not in ages
```

```
True
```



Adding Elements to Dictionary

- Can add pairs to a dictionary:

`phone = {}`

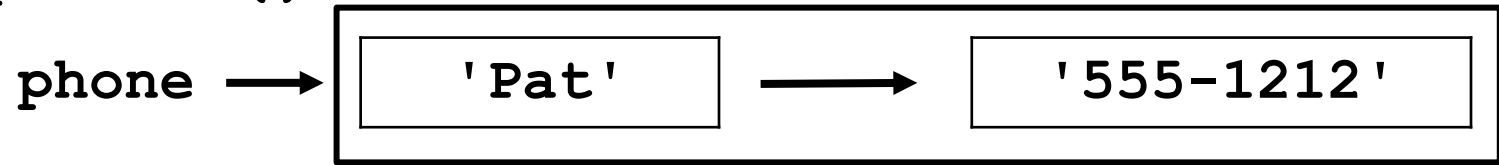
`phone` $\longrightarrow$ *Empty dictionary*



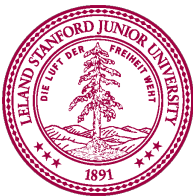
Adding Elements to Dictionary

- Can add pairs to a dictionary:

```
phone = {}
```



```
phone['Pat'] = '555-1212'
```

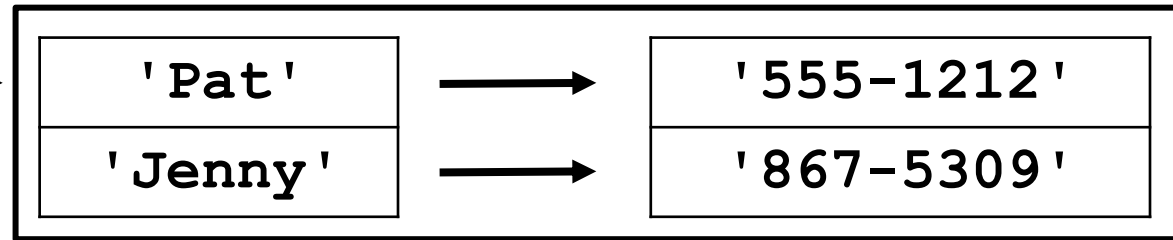


Adding Elements to Dictionary

- Can add pairs to a dictionary:

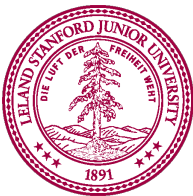
```
phone = {}
```

phone →



```
phone['Pat'] = '555-1212'
```

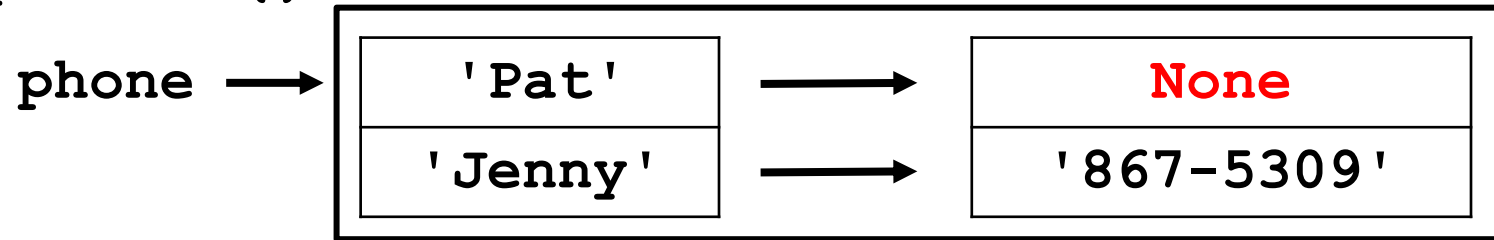
```
phone['Jenny'] = '867-5309'
```



Adding Elements to Dictionary

- Can add pairs to a dictionary:

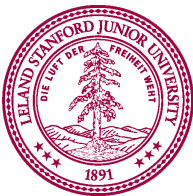
```
phone = {}
```



```
phone['Pat'] = '555-1212'
```

```
phone['Jenny'] = '867-5309'
```

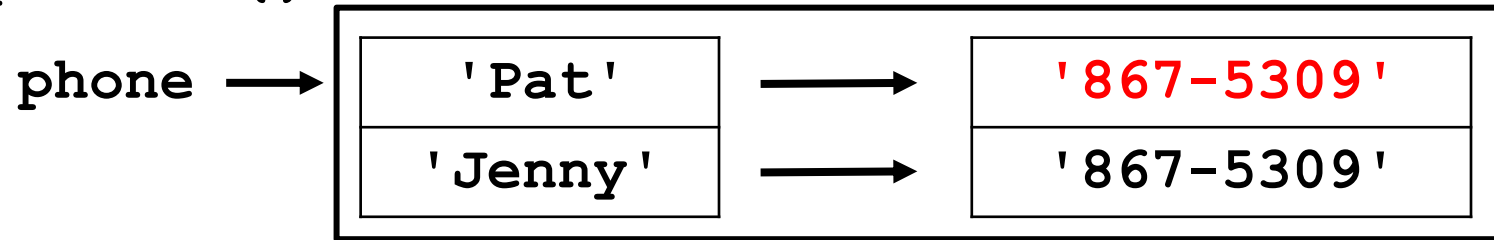
```
phone['Pat'] = None
```



Adding Elements to Dictionary

- Can add pairs to a dictionary:

```
phone = {}
```

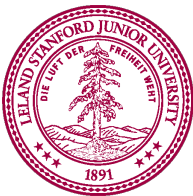


```
phone['Pat'] = '555-1212'
```

```
phone['Jenny'] = '867-5309'
```

```
phone['Pat'] = None
```

```
phone['Pat'] = '867-5309'
```



Looping Over Dictionary

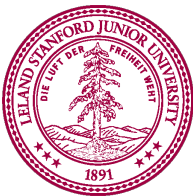
```
ages = {'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

- Can also loop over a dictionary using for-each loop just using name of dictionary:

```
for key in ages:  
    value = ages[key]  
    print(str(key) + ' -> ' + str(value))
```

Terminal:

```
Chris -> 32  
Juliette -> 24  
Mehran -> 50
```



Lets Play!



“Mehran Sahami, as a pixar character, reading a dictionary”

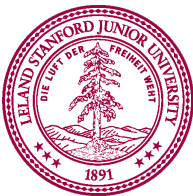
Piech + Sahami, CS106A, Stanford University



Lets Play!



“Professor Mehran Sahami, as a pixar character, reading a dictionary”
Piech + Sahami, CS106A, Stanford University



List

index -> value

Dictionary
key -> value

List

```
my_list = ['a', 'b', 'c']
```

```
print(my_list[1])
```

```
for i in range(len(my_list)):
    value = my_list[i]
    print(i, value)
```

my_list

a	b	c
---	---	---

0

1

2

indices

Dictionary

```
my_dict = {
    'France': '+33',
    'Japan': '+81',
    'Brazil': '+55'
}
```

```
print(my_dict['France'])
```

```
for key in my_dict:
    value = my_dict[key]
    print(key, value)
```

my_dict

+33	+81	+55
-----	-----	-----

'France'

'Japan'

'Brazil'

keys



List

```
my_list = [  
    'a',  
    'b',  
    'c'  
]
```

```
print(my_list[1])
```

```
for i in range(len(my_list)):  
    value = my_list[i]  
    print(i, value)
```

my_list

a	b	c
0	1	2

indices

Dictionary

```
my_dict = {  
    'France': '+33',  
    'Japan': '+81',  
    'Brazil': '+55'  
}
```

```
print(my_dict['France'])
```

```
for key in my_dict:  
    value = my_dict[key]  
    print(key, value)
```

my_dict

+33	+81	+55
'France'	'Japan'	'Brazil'

keys



List

```
my_list = ['a', 'b', 'c']
```

```
print(my_list[1])
```

```
for i in range(len(my_list)):
    value = my_list[i]
    print(i, value)
```

my_list

a	b	c
---	---	---

0

1

2

indices

Dictionary

```
my_dict = {
    'France': '+33',
    'Japan': '+81',
    'Brazil': '+55'
}
```

```
print(my_dict['France'])
```

```
for key in my_dict:
    value = my_dict[key]
    print(key, value)
```

my_dict

+33	+81	+55
-----	-----	-----

'France'

'Japan'

'Brazil'

keys



List

```
my_list = ['a', 'b', 'c']
```

```
print(my_list[1])
```

```
for i in range(len(my_list)):  
    value = my_list[i]  
    print(i, value)
```

my_list

a	b	c
---	---	---

0

1

2

indices

Dictionary

```
my_dict = {  
    'France': '+33',  
    'Japan': '+81',  
    'Brazil': '+55'  
}
```

```
print(my_dict['France'])
```

```
for key in my_dict:  
    value = my_dict[key]  
    print(key, value)
```

my_dict

+33	+81	+55
-----	-----	-----

'France'

'Japan'

'Brazil'

keys



List

```
my_list = ['a', 'b', 'c']
```

```
print(my_list[1])
```

```
for i in range(len(my_list)):
    value = my_list[i]
    print(i, value)
```

my_list

a	b	c
---	---	---

0

1

2

indices

Dictionary

```
my_dict = {
    'France': '+33',
    'Japan': '+81',
    'Brazil': '+55'
}
```

```
print(my_dict['France'])
```

```
for key in my_dict:
    value = my_dict[key]
    print(key, value)
```

my_dict

+33	+81	+55
-----	-----	-----

'France'

'Japan'

'Brazil'

keys



List

```
my_list = ['a', 'b', 'c']
```

```
print(my_list[1])
```

```
for i in range(len(my_list)):
    value = my_list[i]
    print(i, value)
```

my_list

a	b	c
---	---	---

0

1

2

indices

Dictionary

```
my_dict = {
    'France': '+33',
    'Japan': '+81',
    'Brazil': '+55'
}
```

```
print(my_dict['France'])
```

```
for key in my_dict:
    value = my_dict[key]
    print(key, value)
```

my_dict

+33	+81	+55
-----	-----	-----

'France'

'Japan'

'Brazil'

keys



More Dictionary Fun! (Part 1)

```
ages = {'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

- Function: dict.keys()

- Returns something similar to a range of the keys in dictionary
- Can use that to loop over all keys in a dictionary:

```
for key in ages.keys():  
    print(str(key) + " -> " + str(ages[key]))
```

Terminal:

```
Chris -> 32  
Juliette -> 24  
Mehran -> 50
```

- Can turn `keys()` into a list, using the `list` function

```
>>> list(ages.keys())  
['Chris', 'Juliette', 'Mehran']
```



More Dictionary Fun! (Part 2)

```
ages = {'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

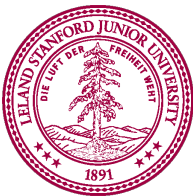
- Function: dict.values()
 - Returns something similar to a range of the values in dictionary
 - Can use that to loop over all keys in a dictionary:

```
for value in ages.values():  
    print(value)
```

Terminal:

```
32  
24  
50
```

- Can turn values() into a list, using the list function
- ```
>>> list(ages.values())
[32, 24, 50]
```





# More Dictionary Fun! (Part 3)

```
ages = {'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

- Function: dict.items()

- Returns
- Can use that to loop over all key value pairs in a dictionary:

```
for key, value in ages.items():
 print(f"{key}, {value}")
```

Terminal:

```
Chris, 32
Juliette, 24
Mehran, 50
```



# More Dictionary Fun! (Part 4/5)

```
ages = {'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

- Function: dict.pop(key)

- Removes key/value pair with the given key. Returns value from that key/value pair.

```
>>> ages
```

```
>>> {'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

```
>>> ages.pop('Mehran')
```

```
50
```

```
>>> ages
```

```
{'Chris': 32, 'Juliette': 24}
```

- Function: dict.clear()

- Removes all key/value pairs in the dictionary.

```
>>> ages.clear()
```

```
>>> ages
```

```
{}
```



# Functions You Can Apply

```
ages = {'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

- Function: `len(dict)`

- Returns number of key/value pairs in the dictionary

```
>>> ages
```

```
{'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

```
>>> len(ages)
```

```
3
```

- Function: `del dict[key]`

- Removes key/value pairs in the dictionary.

- Similar to `pop`, but doesn't return anything.

```
>>> ages
```

```
{'Chris': 32, 'Juliette': 24, 'Mehran': 50}
```

```
>>> del ages['Mehran']
```

```
>>> ages
```

```
{'Chris': 32, 'Juliette': 24}
```



Example:  
count\_nums.py

Example:  
phonebook.py

# Learning Goals

1. Learning about dictionaries
2. Building programs using dictionaries

